Exploring Vortex Stability in Jupiter's Atmosphere Using Linearized Shallow Water Equations

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Abstract

Jupiter's Great Red Spot (GRS) has persisted for centuries, presenting a natural environment for studying long-lived atmospheric vortices in a rapidly rotating environment. This paper investigates the dynamical balance that sustains such structures using a linearized shallow-water framework. The governing equations were nondimensionalized to identify the relative roles of the Froude (Fr), Rossby (Ro), Reynolds (Re), and forcing frequency (ω) parameters. Mean zonal flows derived from Voyager observations by Limaye (1985) provided the background velocity profile, and the equilibrium height was solved under geostrophic balance on a β–plane.

Simulations were performed for a range of nondimensional parameters representing plausible conditions within Jupiter's weather layer. Results show that decreasing Fr enhances overall flow strength without altering structure, while varying Ro transitions the system from rotation-dominated, zonal flow to inertially driven, vortex-intensified regimes. Re primarily affects dissipation and small-scale damping, and low-frequency forcing ($\omega \leq 0.5$) produces resonant amplification of energy consistent with the GRS's persistence. The kinetic-energy and TKE fields reveal strong shear-driven energy exchange near the vortex core, providing insight into how Jupiter's large-scale storms maintain coherence over long timescales.

1 Introduction

Jupiter's Great Red Spot (GRS) is a fascinating phenomenon known for its remarkable longevity, which has been observed for at least 190 years and possibly much longer, though early records may describe a different storm [1]. The exceptional stability of this vortex is unlike anything found in Earth's atmosphere, making the GRS and Jupiter's surrounding bands of high shear ideal for studying long-lived vortical equilibria. Although Jupiter's atmosphere is extremely deep, these cyclonic storms occur within its relatively thin weather layer, where the Shallow-Water Equations (SWE) provide a useful first-order model of large-scale horizontal flow.

Images from Voyager 1 and 2 enabled researchers to estimate zonal wind speeds within 60 degrees latitude from the equator. Sanjay S. Limaye (1985) synthesized these measurements, resolving the structure of Jupiter's easterly and westerly jets [2]. His data provide the mean zonal flow profile $U(y)$ used as the background velocity in this study.

With that data, this paper analyzes the linear response of the GRS and nearby shear bands to small perturbations. The model includes Coriolis and dissipative effects and introduces a height-forcing term that perturbs layer thickness, mimicking localized pressure or density anomalies. In a rapidly rotating atmosphere, such height perturbations naturally produce velocity responses through geostrophic balance, generating vortices without directly imposing momentum [3].

Three nondimensional parameters characterize the relative strength of forces in the system. The Froude number ($Fr = U_0/\sqrt{gH_0}$) compares inertia to gravity and indicates how strongly height variations affect motion. The Rossby number ($Ro = U_0/fL$) measures the ratio of inertial to Coriolis forces, determining how rotation constrains the flow. The Reynolds number ($Re = U_0 L/\nu$) quantifies the competition between inertia and viscosity, setting the degree of dissipation. Together, these quantities reveal the balance of physical processes that govern planetary-scale vortices.

This work extends earlier shallow-water studies of the Great Red Spot [4] by applying Limaye's observed zonal flow profile and systematically varying these nondimensional parameters to examine how they control vortex stability and energy retention.

2 Methods

This section outlines the derivation of the nondimensional, linearized shallow-water equations (SWE), the processing of the observational data used to construct the background flow, and the procedures for numerical analysis and visualization.

2.1 Calculations and Derivations

We begin with the dimensional shallow-water equations (SWE), which describe the evolution of a horizontally moving fluid layer whose thickness is small compared to its horizontal extent. In this system, $u(x, y, t)$ and $v(x, y, t)$ denote the horizontal velocity components in the longitudinal (x) and latitudinal (y) directions, $\eta(x, y, t)$ represents the free-surface displacement relative to a reference depth, and $h = \bar{h} + \eta$ is the total fluid thickness.

The dimensional form of the equations, including Coriolis, viscous, and external forcing terms, is given by [5]:

$$u_t + uu_x + vu_y - fv + g\,\eta_x - \nu\nabla^2 u = \hat{f}_u, \tag{1}$$

$$v_t + uv_x + vv_y + fu + g\,\eta_y - \nu\nabla^2 v = \hat{f}_v, \tag{2}$$

$$\eta_t + \partial_x[(\bar{h} + \eta)u] + \partial_y[(\bar{h} + \eta)v] = \hat{f}_\eta. \tag{3}$$

The first two equations, often referred to as the momentum equations, describe the horizontal acceleration of the fluid. The terms uu_x and vu_y (or uv_x and vv_y in the second equation) represent nonlinear advection of momentum by the flow itself. The terms $-fv$ and $+fu$ are the Coriolis terms, accounting for the deflection of moving air parcels due to Jupiter's rapid rotation, where $f = 2\Omega \sin\phi$ is the Coriolis parameter. The $g\,\eta_x$ and $g\,\eta_y$ terms represent the pressure-gradient forces caused by variations in the surface height η, while the $-\nu\nabla^2 u$ and $-\nu\nabla^2 v$ terms introduce viscous dissipation through the kinematic viscosity ν. The forcing terms $\hat{f}_u$ and $\hat{f}_v$ are external accelerations (e.g., height forcing) that can drive perturbations in the flow.

The third equation is the mass continuity equation, expressing conservation of mass for the shallow layer. The first term, η_t, represents the local rate of change in the free-surface height. The divergence terms, $\partial_x[(\bar{h}+\eta)u]$ and $\partial_y[(\bar{h}+\eta)v]$, describe horizontal inflow or outflow that changes local layer thickness. The right-hand side $\hat{f}_\eta$ allows for external forcing of the fluid thickness, such as the addition or removal of mass or an applied height perturbation.

Together, these equations form a coupled nonlinear system governing horizontal flow and surface deformation in a rotating, viscous layer. These equations serve as the foundation for the nondimensional and linearized forms we will now derive.

We choose a horizontal length scale L (e.g., band width / jet half–width), a reference velocity U_0 (set by the background flow magnitude, and a reference layer depth H_0 (weather–layer equivalent depth). Using

$$x = Lx^*, \quad y = Ly^*, \quad t = \frac{L}{U_0}t^*, \quad u = U_0u^*,\ v = U_0v^*,\ \eta = H_0\eta^*,\ \bar{h} = H_0\bar{h}^*,$$

and scaled forcings

$$\hat{F}_u = \frac{L}{U_0^2}\hat{f}_u, \qquad \hat{F}_v = \frac{L}{U_0^2}\hat{f}_v, \qquad \hat{F}_\eta = \frac{L}{H_0U_0}\hat{f}_\eta,$$

we divide the momentum equations by U_0^2/L and the mass equation by H_0U_0/L to obtain the intermediate nondimensional system (dropping asterisks):

$$u_t + u\,u_x + v\,u_y \;-\; \underbrace{\left(\frac{fL}{U_0}\right)}_{\text{rotation}} v \;+\; \underbrace{\left(\frac{gH_0}{U_0^2}\right)}_{\text{gravity}} \eta_x \;-\; \underbrace{\left(\frac{\nu}{U_0L}\right)}_{\text{viscosity}} \nabla^2 u = \hat{F}_u, \tag{4}$$

$$v_t + u\,v_x + v\,v_y \;+\; \left(\frac{fL}{U_0}\right) u \;+\; \left(\frac{gH_0}{U_0^2}\right) \eta_y \;-\; \left(\frac{\nu}{U_0L}\right) \nabla^2 v = \hat{F}_v, \tag{5}$$

$$\eta_t \;+\; \partial_x\big[(\bar{h}+\eta)u\big] \;+\; \partial_y\big[(\bar{h}+\eta)v\big] = \hat{F}_\eta. \tag{6}$$

Introducing the usual nondimensional numbers

$$Ro \;=\; \frac{U_0}{fL}, \qquad Fr \;=\; \frac{U_0}{\sqrt{gH_0}}, \qquad Re \;=\; \frac{U_0L}{\nu},$$

the prefactors in (4)–(5) become

$$\frac{fL}{U_0} = \frac{1}{Ro}, \qquad \frac{gH_0}{U_0^2} = \frac{1}{Fr^2}, \qquad \frac{\nu}{U_0L} = \frac{1}{Re},$$

yielding the compact nondimensional form used throughout:

$$u_t + u\,u_x + v\,u_y - \frac{1}{Ro}\,v + \frac{1}{Fr^2}\,\eta_x - \frac{1}{Re}\,\nabla^2 u = \hat{F}_u, \tag{7}$$

$$v_t + u\,v_x + v\,v_y + \frac{1}{Ro}\,u + \frac{1}{Fr^2}\,\eta_y - \frac{1}{Re}\,\nabla^2 v = \hat{F}_v, \tag{8}$$

$$\eta_t + \partial_x\big[(\bar{h}+\eta)u\big] + \partial_y\big[(\bar{h}+\eta)v\big] = \hat{F}_\eta. \tag{9}$$

To analyze the system's linear response, we expand around a zonal background flow $\bar{U}(y)$ and a constant mean thickness $\bar{h}$:

$$u = \bar{U}(y) + u', \quad v = v', \quad \eta = \eta'.$$

Neglecting second-order terms yields

$$u_t + \bar{U}\,u_x + v\,\bar{U}_y - \frac{1}{Ro}v + \frac{1}{Fr^2}\eta_x - \frac{1}{Re}\nabla^2 u = \hat{f}_u, \tag{10}$$

$$v_t + \bar{U}\,v_x + \frac{1}{Ro}u + \frac{1}{Fr^2}\eta_y - \frac{1}{Re}\nabla^2 v = \hat{f}_v, \tag{11}$$

$$\eta_t + \bar{U}\,\eta_x + \bar{h}\,(u_x + v_y) = \hat{f}_\eta. \tag{12}$$

In matrix form, these can be compactly expressed as

$$\partial_t \begin{bmatrix} u \\ v \\ \eta \end{bmatrix} = L \begin{bmatrix} u \\ v \\ \eta \end{bmatrix} + \hat{f},$$

where

$$L = \begin{bmatrix} -\bar{U}\,\partial_x + \frac{1}{Re}\nabla^2 & \frac{1}{Ro} - \bar{U}_y & -\frac{1}{Fr^2}\partial_x \\ -\frac{1}{Ro} & -\bar{U}\,\partial_x + \frac{1}{Re}\nabla^2 & -\frac{1}{Fr^2}\partial_y \\ -\bar{h}\,\partial_x & -\bar{h}\,\partial_y & -\bar{U}\,\partial_x \end{bmatrix}.$$

Assuming a harmonic time dependence $e^{-i\omega t}$, the steady-state response is given by

$$(-i\omega I - L)\tilde{\mathbf{q}} = \hat{f}, \qquad \tilde{\mathbf{q}} = (-i\omega I - L)^{-1}\hat{f}.$$

The system of linearized equations was then programmed and solved numerically using Julia's built-in linear algebra and differential equation libraries.

2.2 Data Preparation and Model Setup

The background zonal velocity $\bar{U}(y)$ was constructed from the mean flow observations of Limaye (1985), which provide longitudinally averaged velocities between $\pm 60°$ latitude. To nondimensionalize the dataset, velocities were divided by 100, bringing the original range (up to $\sim$150 m/s) to dimensionless magnitudes of 0–1.5. Latitude coordinates were converted from degrees to radians to ensure consistency with the governing equations.

The meridional background velocity was set to $\bar{V} = 0$, consistent with the dominance of zonal motion. On a β–plane, where $f = f_0 + \beta y$, the equilibrium thickness $\bar{h}(y)$ follows from geostrophic balance:

$$\frac{d\bar{h}}{dy} = -\frac{Fr^2}{Ro}\,\beta\,y\,\bar{U}(y). \tag{13}$$

Here, $\beta = 2\Omega\cos\phi/a$ was evaluated at the latitude of the Great Red Spot ($\phi \approx -20°$), giving $\beta \approx 4.3 \times 10^{-12}\,\mathrm{m}^{-1}\mathrm{s}^{-1}$. The resulting $\bar{h}(y)$ profile was computed numerically by integrating this relation with $\bar{h}(0) = 0$ as the reference height.

2.3 Analysis and Diagnostics

Each simulation produced the perturbation fields (u, v, η), from which three key diagnostic quantities were calculated:

$$K = \tfrac{1}{2}(u^2 + v^2), \qquad D = -\tfrac{1}{Re}(u\nabla^2 u + v\nabla^2 v), \qquad P = u\,v\,\bar{U}_y.$$

Here, K represents the *kinetic energy density* of the perturbations, quantifying the total motion contained in the horizontal velocity components. D denotes the *rate of viscous dissipation*, which measures how much of that kinetic energy is lost to frictional or small-scale viscous effects; its magnitude increases with stronger gradients and lower Reynolds number. The final quantity, P, is the *turbulent kinetic energy (TKE) production* by mean shear, indicating the rate at which energy is transferred between the background flow $\bar{U}(y)$ and the perturbations. Positive P corresponds to regions where the mean shear injects energy into the perturbations, while negative P indicates local energy loss back to the mean flow. Together, these diagnostics provide a complete picture of how perturbations gain, lose, and redistribute energy across the simulated flow field.

Parameter sweeps were performed to explore a range of dynamical regimes:

$$\begin{aligned} &Fr \in [0.5, 1.0, 1.5, 2.0], \quad Ro \in [0.5, 1.0, 1.25, 1.5, 1.75, 2.0], \\ &Re \in [500, 1000, 5000], \quad \omega \in [0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0]. \end{aligned}$$

Simulations were run on a two-dimensional grid with periodic boundaries in x (longitude) and open boundaries in y (latitude). Outputs were visualized using Julia's PyPlot package, producing contour and quiver plots of velocity structure, kinetic energy, dissipation, and TKE production.

3 Results

3.1 Overview of the Great Red Spot Band

Among the analyzed latitude bands, the region between -25° and -15° degrees latitude, corresponding to the Great Red Spot (GRS), exhibited the most prominent vortex structures and strongest zonal shear. The following results therefore focus on this region as a representative case for the broader atmospheric dynamics. Other latitude bands (17°-32° and 40°-55°) were analyzed for consistency, showing qualitatively similar results and are summarized in Appendix A.

3.2 Baseline Flow and Equilibrium Structure

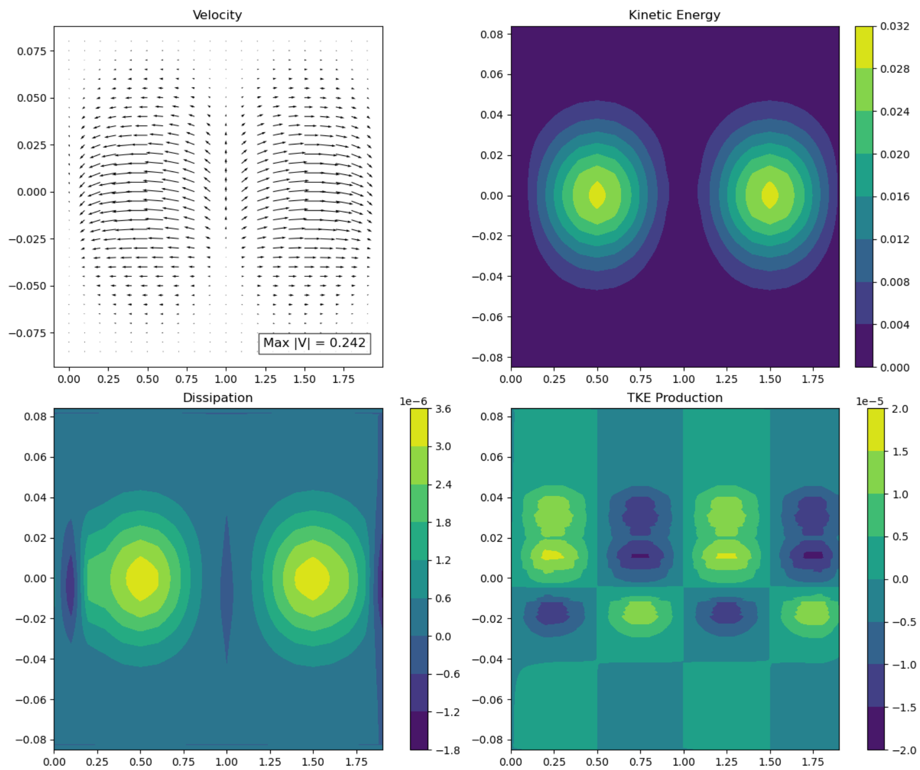
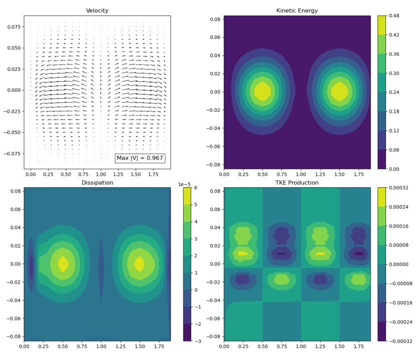


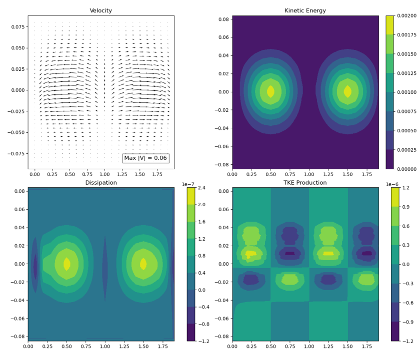
Figure 1: Baseline flow and equilibrium structure for the Great Red Spot band ($Fr = 1.0$, $Ro = 1.0$, $Re = 1000$, $\omega = 2.0$). The plot shows velocity vectors (top left), kinetic energy (top right), dissipation (bottom left), and TKE production (bottom right). *Graphic created by the student researcher using Julia, 2025.*

The baseline simulation corresponds to nondimensional parameters $Fr = 1.0$, $Ro = 1.0$, $Re = 1000$, and $\omega = 2.0$, representing the reference equilibrium for the Great Red Spot (GRS) band spanning latitudes $-25°$ to $-15°$. Figure 1 shows the resulting velocity quiver plot and associated energy fields. The flow exhibits a central longitudinal velocity core near the midpoint of the domain, with outward divergence toward the flanks and weaker, recirculating flow near the edges. The kinetic energy and dissipation fields show two ovals of intensified motion to the left and right of the central flow, corresponding to regions of strong shear and enhanced turbulent energy production.

3.3 Effect of Froude Number (Fr)



(a) $Fr = 0.5$

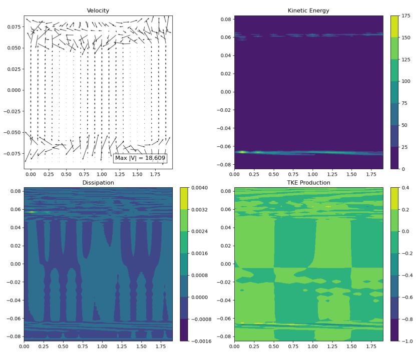


(b) $Fr = 2.0$

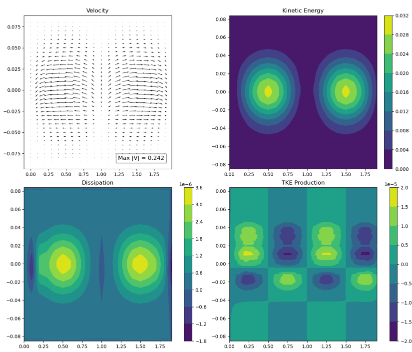
Figure 2: Variation of the equilibrium structure with Froude number. Lower Fr values produce stronger flow intensity but similar spatial patterns.

Varying the Froude number changes the relative influence of gravity on the flow. As shown in Figure 2, decreasing Fr increases the strength of the flow and the amplitude of kinetic energy, while the overall flow structure remains stable. This occurs because a smaller Froude number corresponds to a shallower equilibrium height, which amplifies the effective height gradients and strengthens the driving pressure forces. However, the spatial distribution of kinetic energy and dissipation remains nearly unchanged, implying that gravity primarily scales intensity rather than flow geometry.

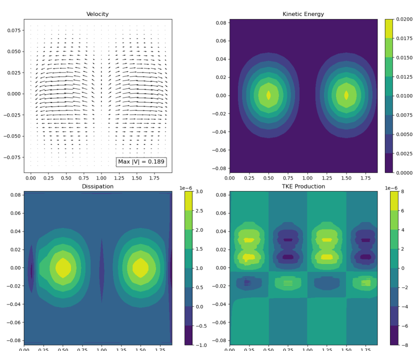
3.4 Effect of Rossby Number (Ro)



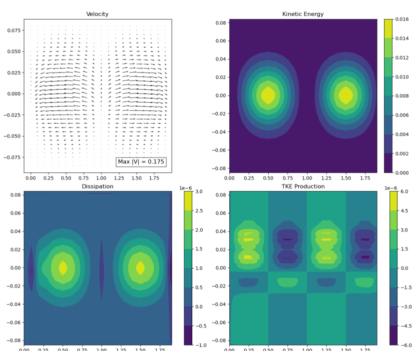
(a) $Ro = 0.5$



(b) $Ro = 1.0$



(c) $Ro = 1.5$

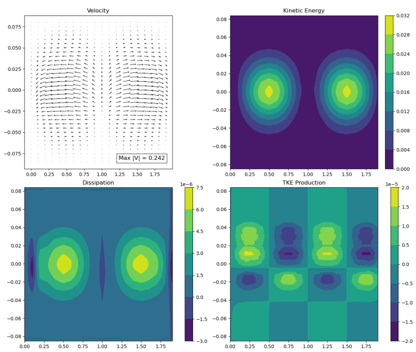


(d) $Ro = 2.0$

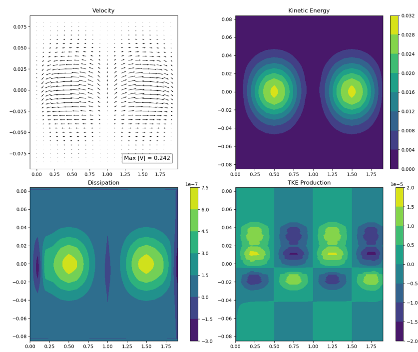
Figure 3: Variation of flow and energy structure with Rossby number for the Great Red Spot band. Lower Ro values produce zonally aligned flows dominated by Coriolis forces, while higher Ro values promote stronger central vortices and localized TKE amplification. *Graphic created by the student researcher using Julia, 2025.*

The Rossby number controls the balance between inertial and Coriolis forces. At low Ro, rotation dominates, producing zonally aligned flows and wide east–west bands, while at higher Ro, inertial effects strengthen, allowing the flow to bend more freely in the latitudinal direction. As Ro increases, the velocity field becomes more concentrated near the center of the band, and TKE production rises sharply, reflecting stronger shear and localized turbulence. This progression illustrates the transition from a rotation-dominated, stable regime to one where inertial instabilities begin to form near the core of the vortex.

3.5 Effect of Reynolds Number (Re)



(a) $Re = 500$

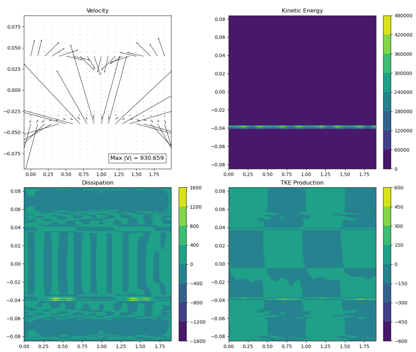


(b) $Re = 5000$

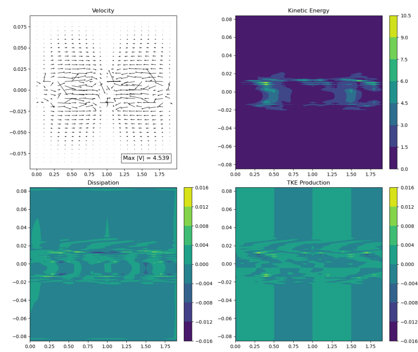
Figure 4: Effect of Reynolds number on the equilibrium structure. Lower Re increases viscous damping and smooths small-scale structures, while higher Re preserves sharper gradients and more localized features. *Graphic created by the student researcher using Julia, 2025.*

The Reynolds number determines the relative importance of viscous damping compared to inertial forces in the flow. As Re decreases, viscous effects strengthen, resulting in higher energy dissipation, smoother velocity fields, and the suppression of small-scale structures. At higher Re, viscous damping weakens, allowing the development of sharper gradients and more localized energy concentrations around the edges of the primary vortex. However, even at very high Re, the overall structure of the vortex remains stable, indicating that Jupiter's weather layer, with its naturally high Reynolds number, is largely inviscid at the scales relevant to the Great Red Spot.

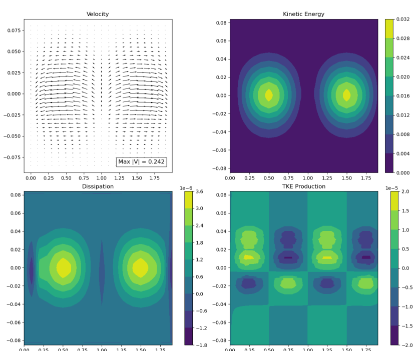
3.6 Effect of Forcing Frequency (ω)



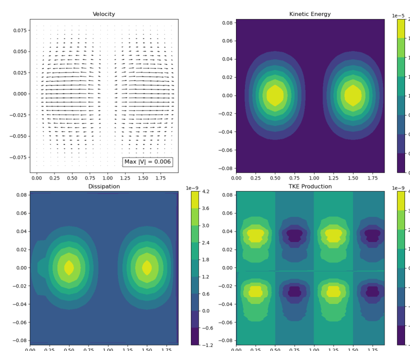
(a) $\omega = 0.1$



(b) $\omega = 0.5$



(c) $\omega = 2.0$



(d) $\omega = 10.0$

Figure 5: Response of the Great Red Spot band to variations in forcing frequency ω. Low-frequency forcing ($\omega \leq 0.5$) produces near-resonant energy amplification, while high-frequency forcing ($\omega \geq 5$) leads to weaker, incoherent flow structures. *Graphic created by the student researcher using Julia, 2025.*

The forcing frequency ω determines how quickly perturbations oscillate over time. When ω is small, perturbations vary slowly enough for the system to respond quasi-steadily, allowing energy to accumulate near resonance. This results in large-amplitude velocities and intensified TKE near the vortex core. As ω increases, the perturbations change too rapidly for the flow to adjust, leading to lower energy, smoother patterns, and weakened vortex coherence. The range of frequencies tested demonstrates this progression from near-resonant amplification at low ω to heavily damped, non-resonant behavior at high ω.

4 Discussion

The most common flow configuration in our simulations (Fig. 1) exhibits a central longitudinal velocity core that curls outward toward both sides of the domain. The flow then accelerates horizontally toward the outer longitudes before converging near the edges and reversing direction, forming a closed circulation. The kinetic energy field shows distinct ovals of enhanced energy flanking the center, while dissipation follows a nearly identical pattern, indicating that energy loss is concentrated in the same high-shear regions. The turbulent kinetic energy (TKE) production field (bottom right of Fig. 1) shows a checkerboard pattern of alternating positive and negative zones, representing the intermittent transfer of energy between the background shear and perturbations near the core of the vortex.

As shown in Figure 2, varying the Froude number (Fr) alters the overall strength but not the structure of the flow. A smaller Fr increases the flow velocity and kinetic energy, reflecting the stronger gravitational contribution to the pressure gradient force. Because Fr directly scales the equilibrium height in the shallow-water model, a lower Fr leads to larger effective height gradients, which in turn produce stronger accelerations. Dissipation and TKE production scale proportionally, indicating that gravity primarily modulates the intensity of the motion rather than its geometry.

The effect of the Rossby number (Ro) is most clearly seen in Figure 3. At low Ro values (subfigure 3a), the Coriolis effect dominates, forcing the flow into broad east–west bands that suppress strong vertical motion. As Ro increases (subfigures 3c–d), the Coriolis constraint weakens, and inertial effects allow the flow to bend more freely in the meridional direction. This results in a concentrated central jet and increased TKE production within the shear region of the vortex, consistent with an energy transfer from the mean flow to the perturbations. The evolution across the four panels of Fig. 3 illustrates the transition from a rotation-dominated regime to one characterized by localized inertial instability near the vortex core.

Figure 4 demonstrates that the Reynolds number (Re) primarily controls dissipation rather than flow organization. At low Re, viscous diffusion smooths gradients, damping small-scale motion and reducing the amplitude of kinetic energy. At high Re, dissipation weakens and sharper gradients form, allowing more persistent shear structures near the periphery of the vortex. Despite these variations, the overall flow topology remains similar, supporting the interpretation that Jupiter's weather layer, with its very high natural Reynolds number, is effectively inviscid at planetary scales.

The dependence on forcing frequency ω (Fig. 5) shows how the temporal scale of perturbations governs energy amplification. At low frequencies ($\omega \leq 0.5$, subfigures 5a–b), the system responds quasi-steadily to the forcing, allowing kinetic and turbulent energy to build up near resonance. At certain frequencies, this resonance becomes particularly strong, producing a rapid increase in energy that exceeds the magnitudes seen in other regimes. These high-energy cases likely represent idealized or non-physical growth arising from the absence of nonlinear damping in the linearized model, but they may also point to the same resonance processes that enable extremely strong, long-lived vortices like Jupiter's Great Red Spot. As ω increases ($\omega \geq 5$, subfigure 5d), the forcing becomes too rapid for the flow to adjust, and the response weakens as the vortex loses coherence. Overall, the progression across Figure 5 illustrates how near-resonant forcing can amplify motion within the GRS core, suggesting that slow, periodic perturbations could play a role in maintaining its remarkable energy and stability.

Overall, these results reveal how the interplay between Coriolis effects, gravitational forcing, viscous damping, and temporal forcing shapes the dynamics of Jupiter's Great Red Spot. Among the parameters tested, Fr and Ro most directly influence the magnitude and organization of the flow, while Re and ω govern its stability and energy balance. The checkerboard TKE patterns and persistent high-energy cores suggest that weakly forced, high-Reynolds-number flows with moderate Coriolis influence can sustain long-lived vortex structures analogous to those seen in Jupiter's atmosphere.

5 Conclusion

This study used a linearized shallow-water model to analyze the dynamics of Jupiter's Great Red Spot and surrounding zonal band. By systematically varying the nondimensional parameters Fr, Ro, Re, and ω, we identified how gravity, rotation, viscosity, and temporal forcing interact to sustain large-scale vortices in a rapidly rotating atmosphere. The results indicate that moderate Rossby numbers and low forcing frequencies produce the most stable, energy-retaining structures, consistent with the long-lived nature of the Great Red Spot. Meanwhile, Reynolds number primarily controls small-scale damping, and Froude number governs the overall intensity of the flow through its scaling of the equilibrium height.

Although the linearized approach captures first-order responses to perturbations, it does not include nonlinear feedbacks that likely contribute to the long-term evolution and energy maintenance of Jupiter's vortices. Future work could incorporate nonlinear shallow-water or fully three-dimensional modeling to explore vortex merging, energy cascades, and radiative coupling with deeper layers. Additionally, direct comparison with high-resolution observations from the Juno mission could help validate the model's predicted energy distributions and resonance behavior. These findings provide a foundation for understanding the balance of forces in planetary atmospheres and for extending similar analyses to other gas giants or exoplanetary systems. This framework demonstrates that simplified fluid models can reproduce key dynamical features of giant-planet vortices, offering a bridge between observational data and fundamental geophysical theory.

Acknowledgments

The author thanks mentor Eojin Kim for guidance throughout this research project. Data on Jupiter's zonal flows were obtained from Limaye (1985).

Appendix A: Additional Latitude Bands

To confirm that the results observed in the Great Red Spot (GRS) band are representative of Jupiter's broader atmospheric dynamics, two additional latitude bands were analyzed. The first band, between 17° and 32° latitude (Band B), corresponds to a mid-latitude region with moderate shear, while the second band, between 40° and 55° latitude (Band C), represents a higher-latitude flow regime with weaker mean zonal velocities. In both cases, the same nondimensional parameter ranges (Fr, Ro, Re, and ω) were tested.

The qualitative flow features for Bands B and C are broadly consistent with those in the GRS region. Each shows symmetric outward flow from the center longitude, oval-shaped kinetic energy peaks near the shear boundaries, and checkerboard patterns of TKE production indicating alternating regions of energy input and extraction. However, at higher latitudes, the magnitude of TKE production and overall flow velocity are slightly reduced, consistent with the decrease in mean zonal flow magnitude in Limaye's dataset.

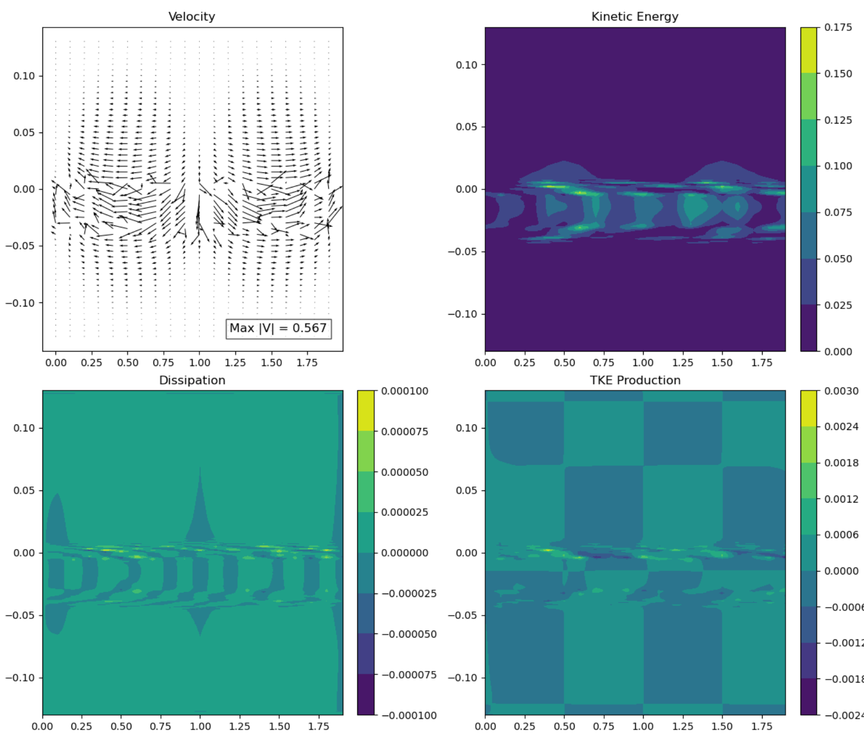


Figure 6: Flow structure and energy fields for Band B (17°–32° latitude) at baseline parameters ($Fr = 1.0$, $Ro = 1.0$, $Re = 1000$, $\omega = 2.0$). *Graphic created by the student researcher using Julia, 2025.*

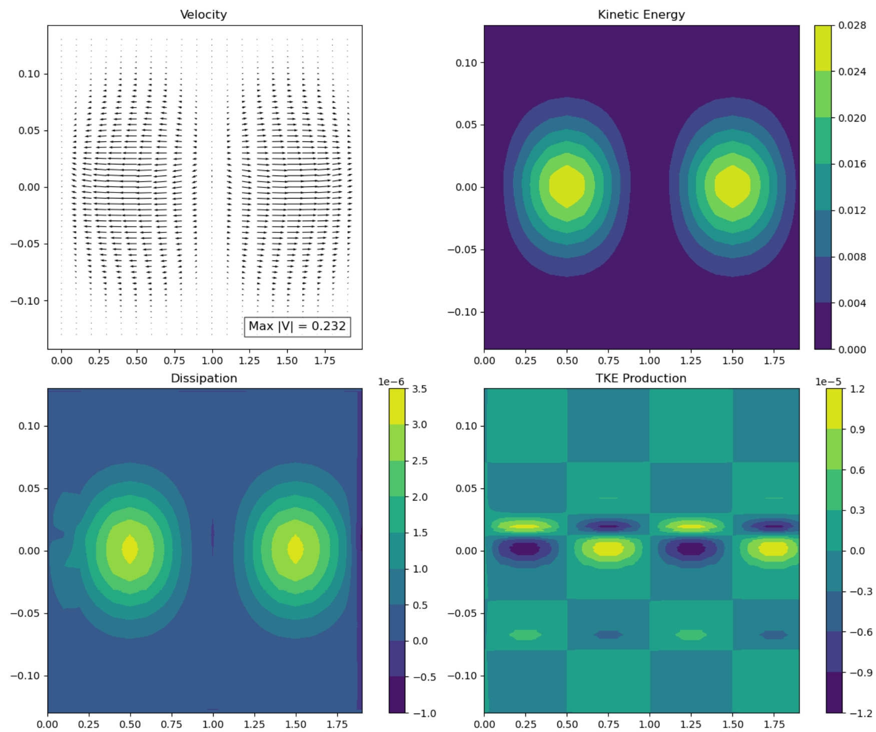


Figure 7: Flow structure and energy fields for Band C (40°–55° latitude) at baseline parameters ($Fr = 1.0$, $Ro = 1.0$, $Re = 1000$, $\omega = 2.0$). *Graphic created by the student researcher using Julia, 2025.*

The similarity in qualitative structure across these three latitude bands supports the robustness of the model. While the absolute energy magnitudes vary with the strength of the local mean flow, the same underlying mechanisms—shear-driven TKE production, outward velocity divergence, and flanking kinetic energy ovals—appear consistently. These results reinforce the conclusion that the observed dynamics of the Great Red Spot are characteristic of the broader atmospheric behavior in Jupiter's weather layer.

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